EXPERIMENT – 9

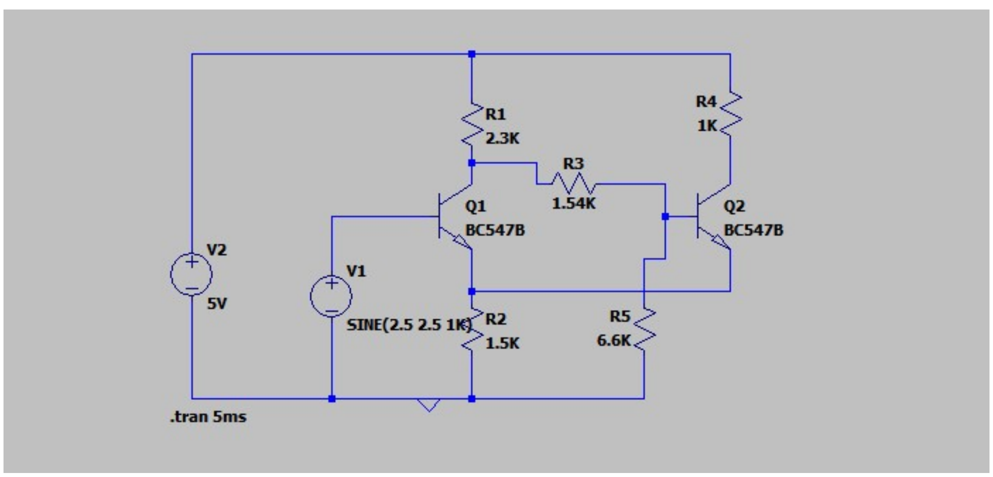
DESIGN AND ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

OBJECTIVES

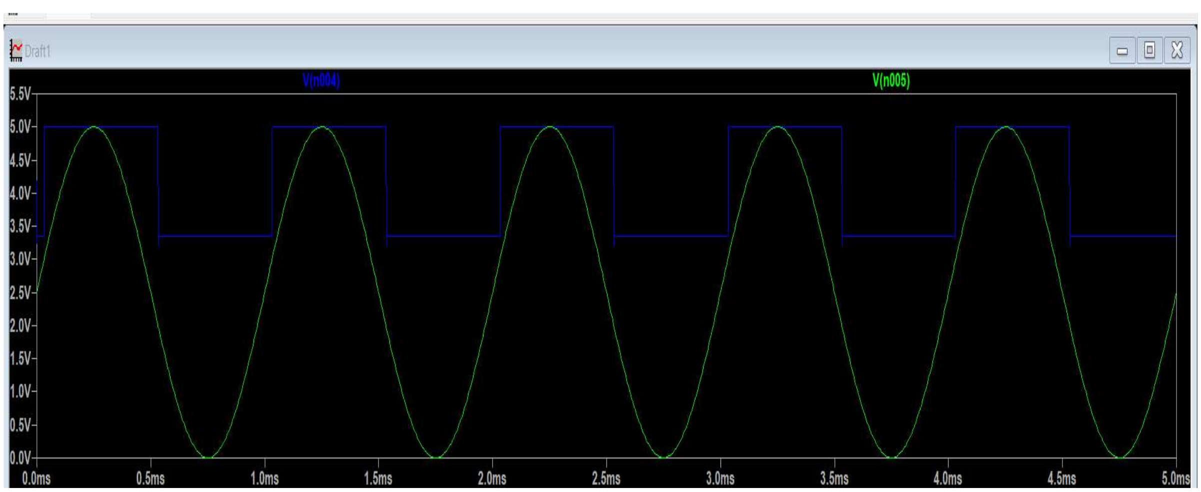
The experiment is divided into three engineering laboratory model tests:

1. Analyse function of Schmitt trigger and determine bandwidth
2. Determination Of UTP and LTP Frequency Response Analysis
3. Analysis of Hysteresis Characteristics

CIRCUIT DIAGRAM:



Model Graph:



EXPERIMENT – 9.1

BANDWIDTH ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

- Understand regenerative feedback.
- Analyze bistable switching action.
- Measure trigger levels.
- Determine bandwidth

Aim

To simulate and verify the operation of a Schmitt Trigger and determine bandwidth using LTSpice.

THEORY

A Schmitt Trigger is a regenerative comparator circuit exhibiting hysteresis.

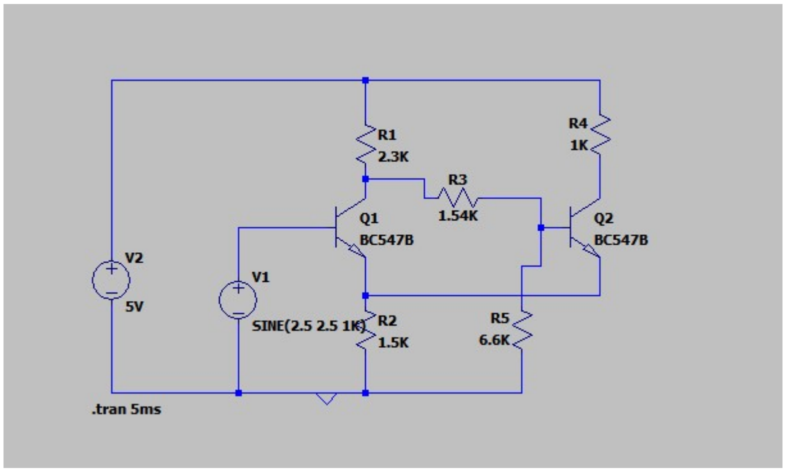
It has:

- Two stable states.
- Two switching thresholds.
- Excellent noise immunity.

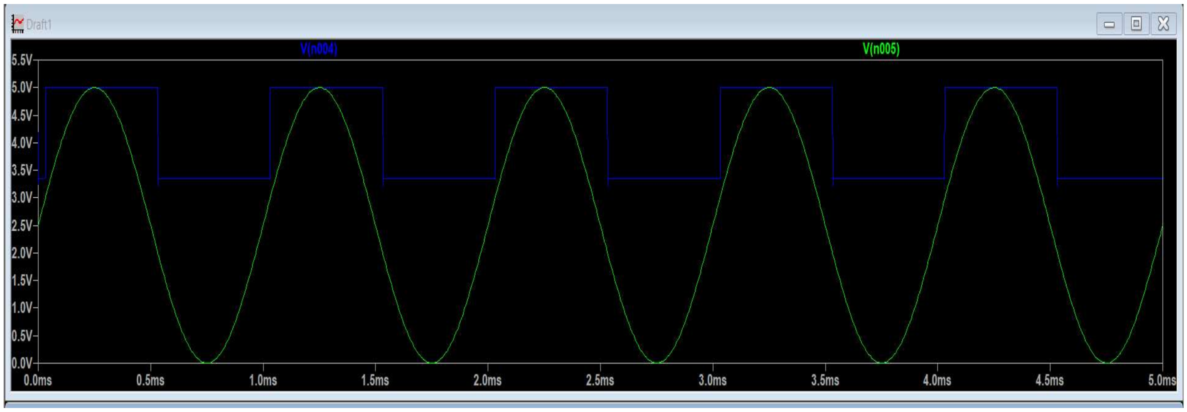
Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

CIRCUIT DIAGRAM:



Model Graph:



BANDWIDTH

PARAMETER	VOUT
AMPLITUDE	1.648 v
TIME PERIOD	1 ms
BANDWIDTH	1.828 Hz

Result:

Schmitt Trigger using LTSpice was completed and output was obtained.

EXPERIMENT – 9.2

DETERMINATION OF UTP AND LTP FOR SCHMITT TRIGGER USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

Aim

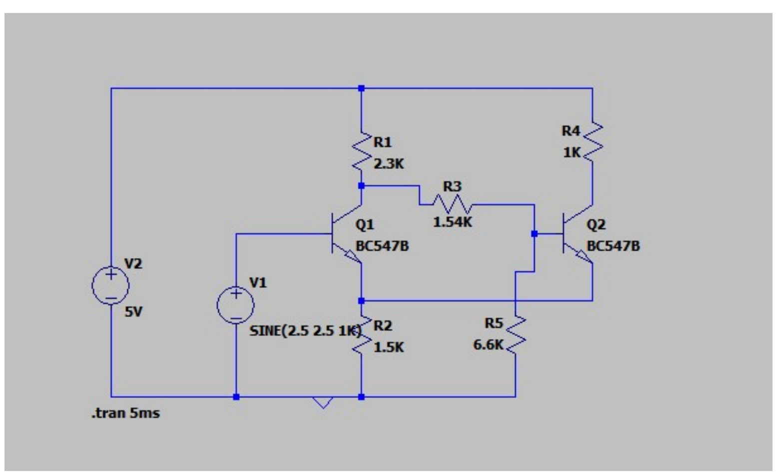
To determine Upper Trigger Point and Lower Trigger Point for Schmit trigger circuit

DESIGN FORMULA:

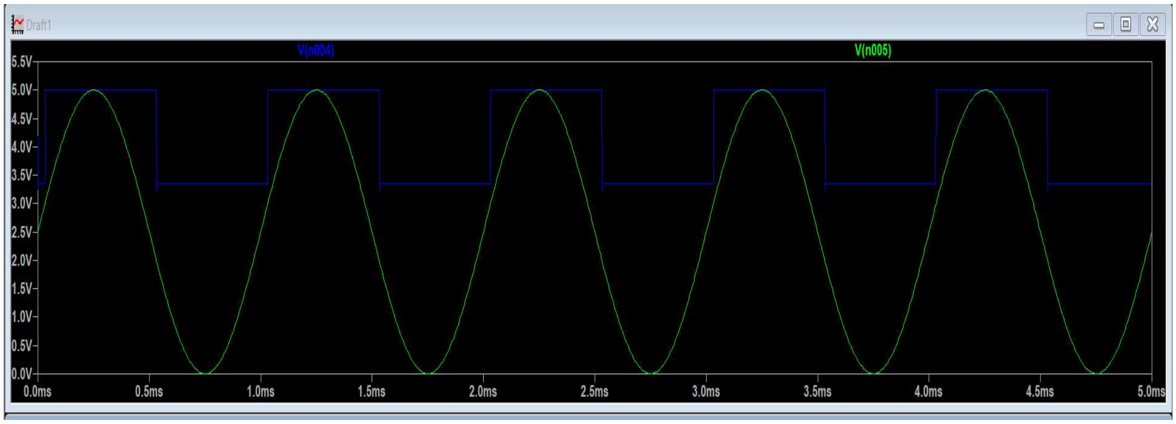
Hysteresis Width

$$V_H = UTP - LTP$$

CIRCUIT DIAGRAM:



Model Graph:



TABULATION

PARAMETER	VOUT
UTP	0.531
LTP	1.028
HYSTERESIS WIDTH	0.497

RESULT

The trigger levels were measured successfully.

EXPERIMENT – 9.3

ANALYSIS OF HYSTERESIS CHARACTERISTICS FOR SCHMITT TRIGGER USING LTSPICE

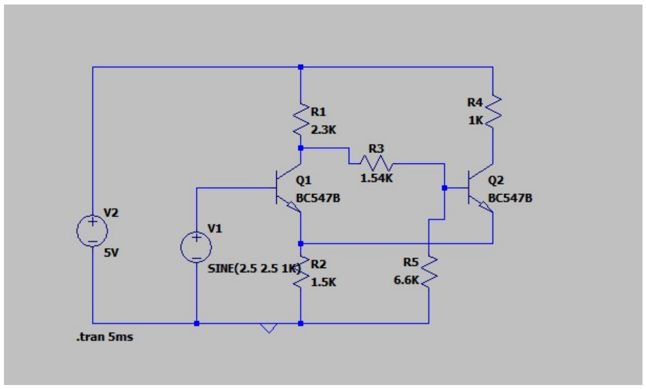
After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Observe and interpret the switching characteristics of the Schmitt Trigger.
7. Plot and analyze the hysteresis transfer characteristic curve.
8. Evaluate the noise immunity provided by hysteresis in comparator circuits.
9. Compare the operation of a Schmitt Trigger with that of a conventional comparator.
10. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

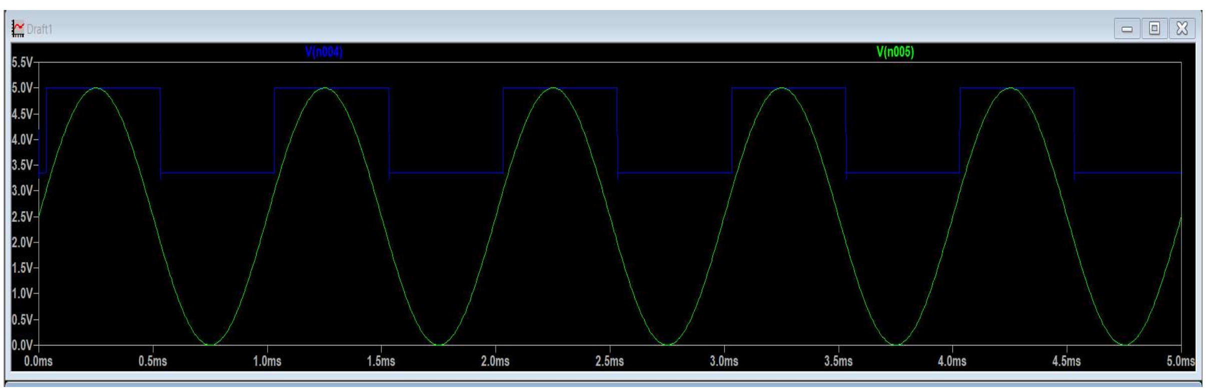
Aim

To plot hysteresis characteristics of a Schmitt Trigger.

CIRCUIT DIAGRAM:



Model Graph:



TABULATION

PARAMETER	Voltage
VIN	5 V
VOUT	4.992 V

RESULT:

Thus the plot hysteresis characteristics of a Schmitt Trigger was done.